

Climate Change and Nutritional Security: Which Crops Are Most Vulnerable?

Mid-evaluation report: a Climate–Nutrition Vulnerability Index for rice, wheat, maize and soybean

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Earth 616 · observed data 1971–2024

Abstract

We build an index of crop vulnerability that combines how strongly a crop’s growing areas are exposed to heat and water stress (*Exposure*), how much yield is lost per unit of that stress (*Sensitivity*), and how much of a population’s energy, protein, fat, zinc and iron the crop supplies (*Importance*). It is applied to rice, wheat, maize and soybean with observed climate from 1971 to 2024, across major producing countries in different climate zones (USA, Brazil, India, China, Russia, South Africa, Bangladesh, Egypt, Nigeria, Argentina). Sensitivity is parameterized directly from established crop literature (heat damage thresholds and FAO yield response factors K_y), providing a standardized per-crop response scale. Wheat emerges as the most vulnerable crop under all candidate index formulas in the main run (Wheat > Maize > Soybean > Rice with equal weights, Wheat > Soybean > Maize > Rice production-weighted) and in 80 of 80 robustness rankings. Wheat leads through high nutritional Importance (supplying the majority of cereal protein, zinc and iron) and substantial climate exposure; maize and soybean are more sensitive to heat per °C but are largely used for feed or processing rather than direct human food. Rice scores lowest because it is predominantly irrigated and monthly flowering-season temperatures remain below its 35 °C critical limit on gridded monthly averages. A standard 10% yield loss reduces the Mean Adequacy Ratio (MAR) most severely where iron supply is already marginal. All figures and tables in this document are generated directly from the analysis pipeline.

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1 Problem statement

Climate change affects food security through two channels: how much of each crop is produced, and what that crop contributes nutritionally. Most published literature addresses only one of these channels: climate explains about a third of global yield variability (Ray et al., 2015), and warming lowers global yields of wheat, rice, maize and soybean by about 6.0, 3.2, 7.4 and 3.1% per °C (Zhao et al., 2017). Elevated CO₂ lowers zinc, iron and protein in C3 crops (Myers et al., 2014; Smith and Myers, 2018), a separate line of research. Because published metrics operate on incompatible scales, no previous study ranks crops by combining climatic exposure, crop sensitivity, and nutritional importance into a unified index.

This report documents our Climate–Nutrition Vulnerability Index (CNVI) framework, its data inputs, calculations, and findings. It addresses two core questions:

1. Which of the four crops is most vulnerable to observed changes in temperature and rainfall, once we also account for dietary importance?
2. If a crop’s yield falls, what happens to a country’s nutrient supply and adequacy?

The framework adapts the IPCC risk structure (Vulnerability = Exposure × Sensitivity × Importance), where *adaptive capacity* is replaced by *nutritional importance*, focusing explicitly on nutritional security at the population level.

2 Data

Dataset	Content	Period	Used for
WorldClim 2.1 historical monthly weather (CRU-TS 4.09 downscaled), 10' (Fick and Hijmans, 2017; Harris et al., 2020)	monthly Tmax, Tmin, precipitation	1970–2024	hazards, Köppen class
SPAM 2010 v2.0 (International Food Policy Research Institute, 2019)	harvested area per crop, irrigated / rainfed, 5'	c. 2010	area weights, rainfed share
Sacks et al. crop calendar (Sacks et al., 2010)	planting and harvest day per cell	climatology	growing and flowering windows
FAOSTAT crops (QCL) (FAO, 2025)	production, area, yield by country	1961–2024	panels, out-of-sample data check
FAOSTAT Food Balances 2010– and –2013	kcal, protein, fat, food quantity per item	2010–23; 1961–2013	Importance, adequacy, pass-through
FAOSTAT food-security indicators	Average Dietary Energy Requirement	2000–23	energy requirement
USDA FoodData Central, SR Legacy (U.S. Department of Agriculture, Agricultural Research Service, 2019)	zinc, iron per 100 g	2018	zinc and iron supply
Natural Earth 1:50 m	borders, continents	static	country masks

Table 1: Datasets used in the analysis pipeline (downloaded by `code/01_download_data.py`; details in `docs/DATA_SOURCES.md`).

Why monthly weather and not normals. A 30-year normal cannot show variability, change or the frequency of bad seasons. Year-by-year data capture observed stress conditions. **Why SPAM and the Sacks calendar.** National averages mix crop areas with unsuitable terrain;

grid-cell crop areas and specific growing calendars ensure climate metrics reflect where and when each crop is actually cultivated. **Food balances.** The 1961–2013 and 2010–2023 FAOSTAT series are analysed separately due to methodological breaks.

3 Methodology

3.1 Country panels

Country panels were selected systematically:

1. **Core countries:** USA, Brazil, India and China for all four crops.
2. **Regional representation:** One European and one African country per crop. Russia represents Europe (ranking in Europe’s top-3 producers for all four crops). In Africa, top producers vary: Nigeria for rice, Egypt for wheat, South Africa for maize and soybean.
3. **World top-3 producers:** Adds Bangladesh for rice and Argentina for soybean.

The panels cover 63% (rice), 51% (wheat), 69% (maize) and 90% (soybean) of global production.

Crop	Country	Rank	Share (%)	Class	Elev. (m)	Crop area in A/B/C/D (%)
Rice	China	1	26.4	C	260	1/1/86/12
	India	2	25.1	A	216	51/12/37/0
	Bangladesh	3	7.4	A	17	62/0/38/0
	Brazil	11	1.4	A	208	53/0/47/0
	USA	13	1.2	C	52	0/0/100/0
	Nigeria	14	1.1	A	368	89/11/0/0
	Russia	39	0.1	C	21	0/10/52/38
Wheat	China	1	17.3	C	386	0/22/51/27
	India	2	13.9	C	255	14/34/52/0
	Russia	3	11.2	D	177	0/1/2/97
	USA	4	6.1	D	647	0/31/24/44
	Egypt	17	1.2	B	42	0/100/0/0
	Brazil	19	1.0	C	567	5/0/95/0
Maize	USA	1	31.0	D	361	0/6/12/83
	China	2	23.3	D	429	0/20/28/52
	Brazil	3	9.2	A	512	53/6/41/0
	India	5	2.9	A	435	37/30/31/1
	S. Africa	8	1.3	B+M	1400	0/52/48/0
	Russia	9	1.2	D	155	0/1/12/87
Soybean	Brazil	1	36.4	A	531	59/0/41/0
	USA	2	31.5	D	278	0/0/22/77
	Argentina	3	11.4	C	159	0/6/94/0
	China	4	5.3	D	333	0/6/33/61
	India	5	3.7	A	420	70/10/20/0
	Russia	8	1.6	D	152	0/0/4/96
	S. Africa	12	0.5	C+M	1273	1/45/55/0

Table 2: Country panels. Rank: global production rank; Share: % of world production; Class: primary Köppen group (+M for mountain elevation ≥ 1000 m).

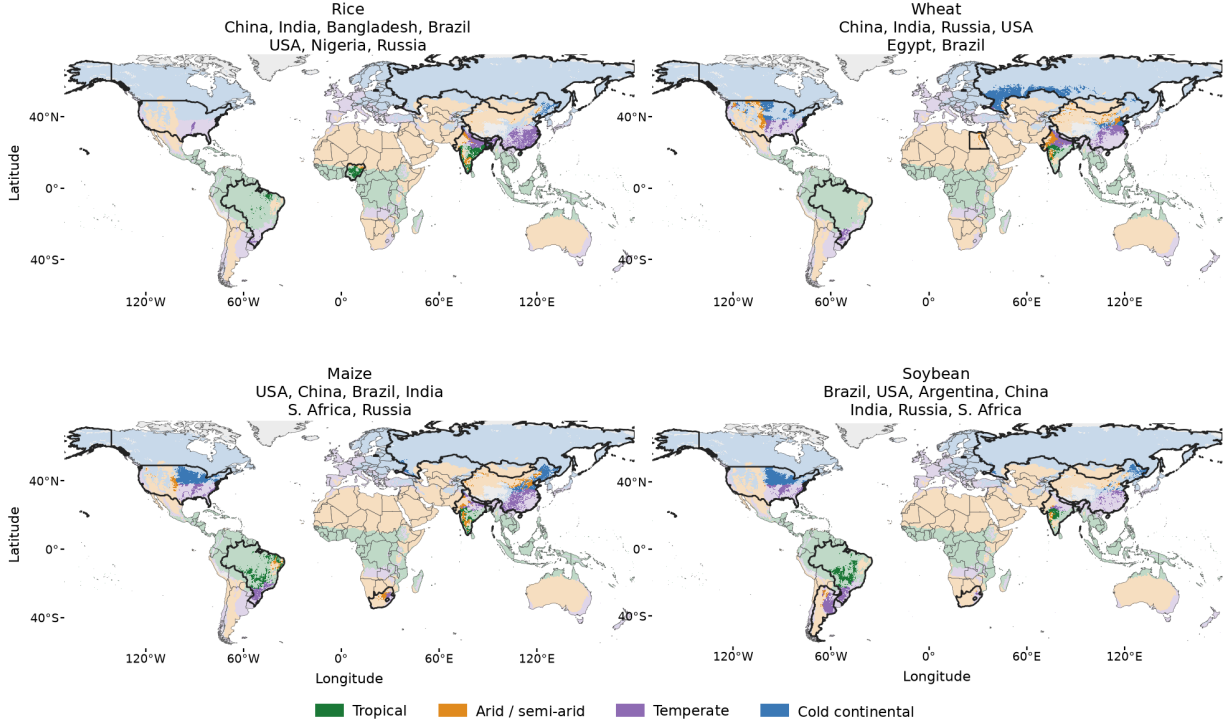


Figure 1: Panel countries (black outlines) and climate classes where each crop grows across Köppen zones (A: tropical, B: arid, C: temperate, D: cold continental).

3.2 Exposure

Exposure measures the physical climate stress experienced by crop growing areas over 2001–2024, evaluated across temperature (heat) and precipitation (water).

Temperature Exposure (E^T). Heat damage concentrates during the reproductive flowering phase (anthesis). Following Deryng et al. (2014), flowering occurs during 45–70% of the growing season length. Heat stress level L^T measures the degrees by which effective temperature $T_{eff} = \frac{1}{2}(T_{mean} + T_{max})$ exceeds the crop critical limit T_{crit} :

$$E^T = \frac{L^T}{\max L^T}, \quad L^T = \text{mean}_{2001-24} \frac{\sum_m w_m \max(0, T_{eff,m} - T_{crit})}{\sum_m w_m} \quad (1)$$

where w_m is the fraction of month m inside the flowering window. Critical limits are: wheat 25 °C, maize 32 °C, soybean 35 °C (Deryng et al., 2014), and rice 35 °C (Yoshida, 1981; Steduto et al., 2012). E^T is normalized to 0–1 across all crop–country pairs by dividing by the maximum observed level ($\max L^T$).

Precipitation Exposure (E^P). Water stress measures the share of crop water requirements unmet by rainfall on rainfed hectares:

$$E^P = \frac{L^P}{\max L^P}, \quad L^P = \text{mean}_{2001-24} \frac{\sum_{cells} R \max(0, 1 - P/ET_c)}{\sum_{cells} A} \quad (2)$$

where P is growing-season precipitation, $ET_c = \bar{K}_c ET_0$ is crop water demand (Allen et al., 1998), R is rainfed harvested hectares, and A is total harvested hectares. Irrigated land is treated as

having no water shortfall ($1 - P/ET_c = 0$). Reference evapotranspiration ET_0 is calculated using the Hargreaves equation (Allen et al., 1998, FAO-56 Eq. 52) with extraterrestrial radiation R_a . E^P is normalized to 0–1 by dividing by $\max L^P$.

3.3 Sensitivity

Sensitivity represents the fraction of yield lost per unit of climate stress, derived directly from established agronomic literature to ensure comparability across crops.

Temperature Sensitivity (S^T). Based on the linear yield loss function of Deryng et al. (2014), yield declines linearly from zero loss at T_{crit} to complete loss at T_{lim} :

$$S^T = \frac{s^T}{\max s^T}, \quad s^T = \frac{1}{T_{lim} - T_{crit}} \quad (3)$$

For rice, $s^T = 0.07$ per °C above $T_{crit} = 35$ °C based on spikelet sterility measurements from Jagadish et al. (2007) and Yoshida (1981). The values are normalized by the maximum slope ($\max s^T = 0.20$ for soybean):

- Soybean: $s^T = 1/(40 - 35) = 0.200 \implies S^T = 1.00$
- Wheat: $s^T = 1/(35 - 25) = 0.100 \implies S^T = 0.50$
- Maize: $s^T = 1/(45 - 32) = 0.077 \implies S^T = 0.38$
- Rice: $s^T = 0.070 \implies S^T = 0.35$

Precipitation Sensitivity (S^P). Based on the FAO yield response factor K_y (Steduto et al., 2012), which relates relative yield loss to relative evapotranspiration deficit:

$$S^P = \frac{K_y}{\max K_y} \quad (4)$$

K_y values are: maize 1.25, wheat 1.10 (average of spring 1.15 and winter 1.05), soybean 0.85, and rice 1.25 (Steduto et al., 2012; Brouwer and Heibloem, 1986). Normalized by $\max K_y = 1.25$:

- Rice & Maize: $K_y = 1.25 \implies S^P = 1.00$
- Wheat: $K_y = 1.10 \implies S^P = 0.88$
- Soybean: $K_y = 0.85 \implies S^P = 0.68$

Crop	Path	T_{crit}/T_{lim} (°C)	s_{heat}^{lit} (per °C)	K_y	K_c ini/mid/end	$\bar{K}_c$
Rice	C3	35 / –	0.070	1.25	1.05/1.20/0.75	1.11
Wheat	C3	25 / 35	0.100	1.10	0.40/1.15/0.33	0.88
Maize	C4	32 / 45	0.077	1.25	0.30/1.20/0.48	0.84
Soybean	C3	35 / 40	0.200	0.85	0.40/1.15/0.50	0.89

Table 3: Literature parameters used for Exposure and Sensitivity. T_{crit}, T_{lim} : critical and limit temperatures; s^T : heat loss slope; K_y : FAO water yield factor; K_c : crop coefficients.

Empirical Yield Regressions (Out-of-Sample Check). Country-level regressions of de-trended yield anomalies on climate variables ($y_t = a + b_T T_{eff,t} + b_W (P/ET_c)_t + \varepsilon_t$ over 1971–2000) are retained exclusively as an empirical validation check against literature benchmarks (e.g. Zhao et al. 2017), rather than being merged into the sensitivity score.

3.4 Importance and Nutrient Adequacy

Dietary Importance (I) measures the population’s nutritional reliance on each crop across five key nutritional dimensions:

$$I_{kc} = \frac{1}{5} \sum_{n \in \{\text{kcal, protein, fat, Zn, Fe}\}} \frac{\text{supply}_n \text{ from crop } k}{\text{total national supply}_n} \quad (2021\text{--}23 \text{ mean}) \quad (5)$$

Calories, protein, and fat come from FAO Food Balance Sheets; zinc and iron are computed by mapping each of the 98 FBS food items to USDA FoodData Central composition values.

Nutrient Adequacy. Adequacy is evaluated via the Nutrient Adequacy Ratio (NAR) and Mean Adequacy Ratio (MAR):

$$\text{NAR}_n = \min \left(1, \frac{\text{supply}_n}{\text{requirement}_n} \right), \quad \text{MAR} = \frac{1}{5} \sum_n \text{NAR}_n \quad (6)$$

Truncating NAR_n at 1 ensures surplus in one nutrient cannot mask deficiency in another. Requirements follow WHO/FAO standards: energy from FAOSTAT Average Dietary Energy Requirement, protein at 10% and fat at 15% of daily energy (WHO/FAO, 2003), zinc at 11.9 mg/day (low bioavailability), and iron at 21.6 mg/day (10% bioavailability) (WHO/FAO, 2004).

3.5 Index Formulation and Candidate Forms

For crop k in country c , four mathematical aggregation formulas were evaluated:

$$\text{F1 Multiplicative: } V = E \cdot S \cdot I, \quad E = \frac{1}{2}(E^T + E^P), \quad S = \frac{1}{2}(S^T + S^P) \quad (7)$$

$$\text{F2 Geometric: } V = (E \cdot S \cdot I)^{1/3} \quad (8)$$

$$\text{F3 Additive: } V = (E + S + I)/3 \quad (9)$$

$$\text{F4 Hazard-Paired: } V = \frac{1}{2}(E^T S^T + E^P S^P) I \quad (10)$$

Formula F4 was chosen for fundamental design reasons:

1. *Hazard-specific coupling:* Each climate hazard multiplies its own corresponding sensitivity ($E^T S^T$ and $E^P S^P$). In F1–F3, high heat exposure can erroneously multiply water sensitivity.
2. *No non-food vulnerability:* If a crop contributes negligibly to direct diet ($I \approx 0$), its nutritional vulnerability remains zero. Additive forms (F3) allow climate exposure alone to produce high vulnerability for non-food crops.
3. *Physical interpretability:* The term $\frac{1}{2}(E^T S^T + E^P S^P)$ corresponds to expected proportional yield loss from combined climate stresses.

4 Results

4.1 Climate Exposure

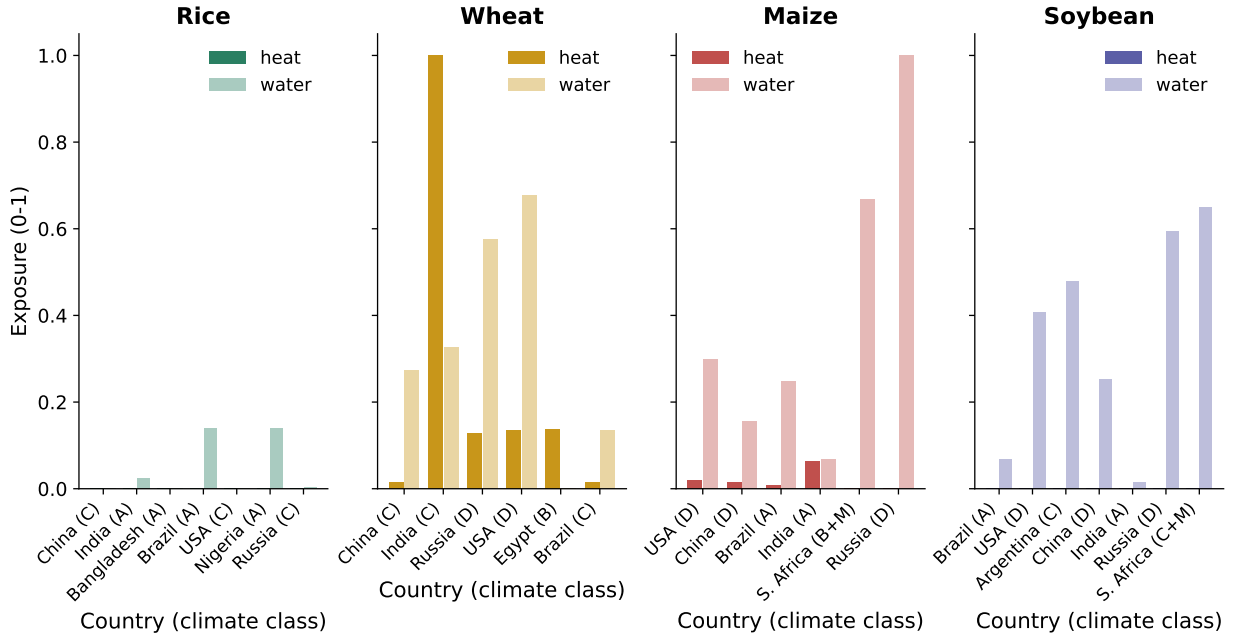


Figure 2: Exposure scores (0–1) by crop and country: temperature exposure E^T (dark) and precipitation exposure E^P (light).

Heat exposure peaks for wheat in India ($E^T = 1.00$) due to hot conditions exceeding 25°C during anthesis. Water exposure is highest for rainfed maize in South Africa and Russia, and wheat in the USA and Russia. Rice exhibits low exposure across both dimensions ($E^T < 0.001$, $E^P < 0.14$) because major paddies are heavily irrigated and monthly averages do not exceed 35°C .

4.2 Nutritional Importance and Adequacy

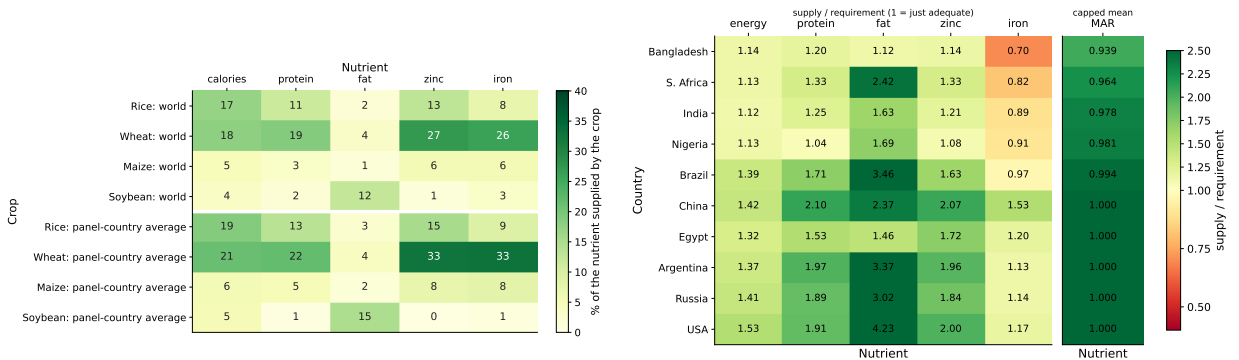


Figure 3: Nutritional supply share (%) by crop.

Figure 4: Nutrient adequacy (supply ÷ requirement).

Wheat and rice provide the bulk of cereal protein, zinc, and iron worldwide (wheat supplies 18.8% of protein, 27.2% of zinc, and 26.0% of iron in panel countries). Soybean contributes primarily fat (12.5%) as most is crushed for oil, while maize contributes modestly to direct human diets (4.9% kcal) as the majority is used for animal feed.

Iron is the binding nutritional constraint: national supply relative to requirement is 0.70 in Bangladesh, 0.82 in South Africa, and 0.89 in India, whereas energy and protein requirements are met at the national level.

4.3 The Vulnerability Index

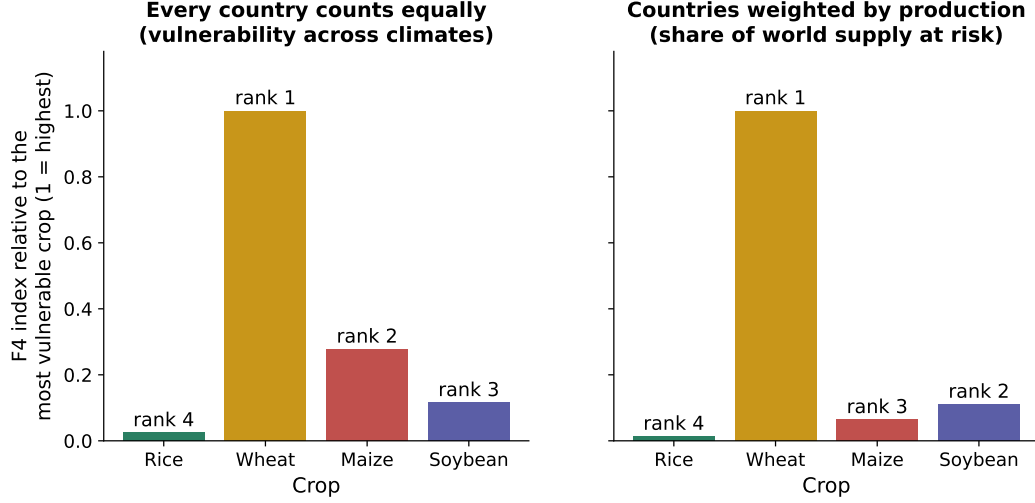


Figure 5: F4 Vulnerability Index relative to the highest-scoring crop (1.0). Left: equal country weights; Right: production-weighted.

Under equal country weights, F4 ranks crop vulnerability as:

Wheat > Maize > Soybean > Rice

Under production weighting, the ordering is:

Wheat > Soybean > Maize > Rice

Form	Country weights	Rice	Wheat	Maize	Soybean
F1 $E \cdot S \cdot I$	equal	0.001 (4)	0.064 (1)	0.021 (2)	0.010 (3)
	production	0.001 (4)	0.081 (1)	0.006 (3)	0.012 (2)
F2 $(ESI)^{1/3}$	equal	0.077 (4)	0.366 (1)	0.201 (2)	0.192 (3)
	production	0.081 (4)	0.407 (1)	0.153 (3)	0.207 (2)
F3 $(E + S + I)/3$	equal	0.229 (4)	0.410 (1)	0.318 (2)	0.297 (3)
	production	0.283 (3)	0.429 (1)	0.242 (4)	0.295 (2)
F4 paired $\times I$	equal	0.002 (4)	0.065 (1)	0.018 (2)	0.008 (3)
	production	0.001 (4)	0.084 (1)	0.005 (3)	0.009 (2)

Table 4: Crop vulnerability index under each candidate formula and weighting (rank in parentheses).

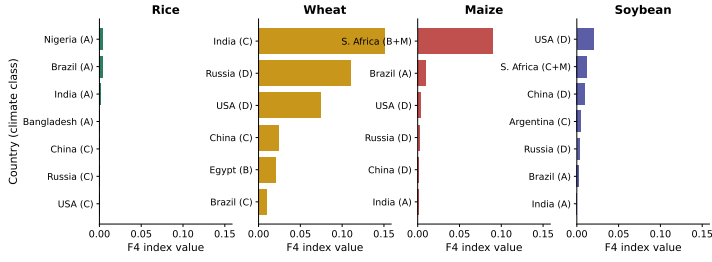


Figure 6: F4 vulnerability index by crop and country.

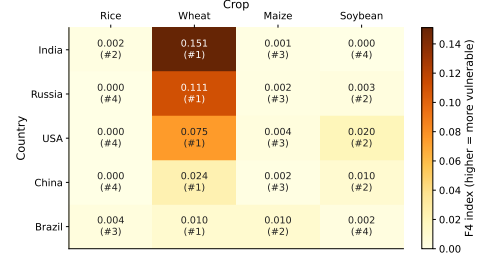


Figure 7: Within-country crop vulnerability rank.

Wheat ranks #1 across all 5 shared panel countries that grow all four crops, driven by the combination of high dietary dependence and substantial climate exposure in major producers (India 0.1511, Russia 0.1105, USA 0.0747). Maize exhibits localized high vulnerability in South Africa (0.0895) where it is a primary dietary staple grown under rainfed conditions.

Country	Rice	Wheat	Maize	Soybean
USA	< 0.0001	0.0747	0.0037	0.0197
Brazil	0.0039	0.0103	0.0103	0.0025
India	0.0023	0.1511	0.0015	0.0002
China	0.0001	0.0240	0.0016	0.0097
Russia	< 0.0001	0.1105	0.0021	0.0032
Argentina	—	—	—	0.0052
Bangladesh	0.0002	—	—	—
Egypt	—	0.0209	—	—
Nigeria	0.0046	—	—	—
South Africa	—	—	0.0895	0.0122

Table 5: Country-by-crop F4 vulnerability index (V_{F4}).

4.4 Robustness and Validation

Robustness. Evaluating 10 one-at-a-time modifications (scaling methods, weighting schemes, parameter proxies) across 4 formulas and 2 weighting types (80 total permutations) confirms that **wheat ranks #1 in 80 of 80 tests**. In Monte Carlo simulations with 2,000 random weight assignments, wheat ranks first in 97.5% of iterations under F4.

Validation. - *Zhao et al. (2017)*: Full F4 index rank correlation with global warming yield declines is $\rho = +0.6$. - *Myers et al. (2014)*: Elevated CO₂ declines disproportionately affect wheat and rice zinc and iron, aligning with our nutritional importance rankings. - *Yield shocks*: Observed 2001–2024 yield shocks show weak positive association ($\rho = +0.20, p = 0.170$) with the climate index.

4.5 Yield to Diet Scenarios

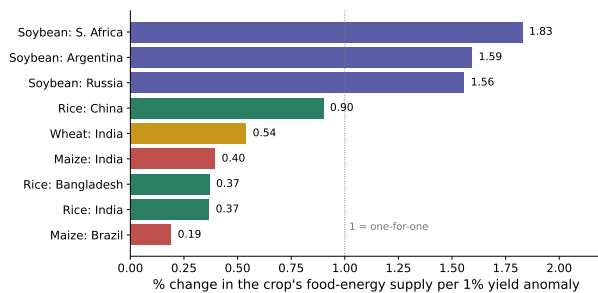


Figure 8: Yield pass-through β to food supply.

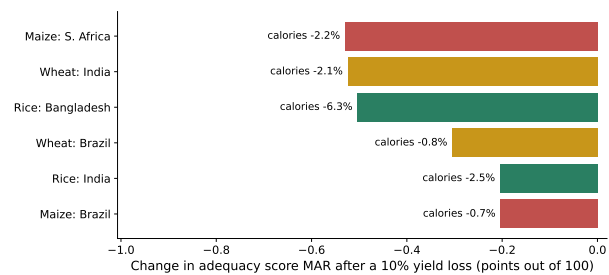


Figure 9: Fall in MAR following a 10% yield loss.

A standard -10% yield loss reduces dietary adequacy most severely for maize in South Africa and wheat in India, where iron availability is already close to requirement levels.

5 Conclusions

1. **Wheat is the most vulnerable crop** to climate-driven nutritional disruption (Wheat > Maize > Soybean > Rice with equal weights; Wheat > Soybean > Maize > Rice production-weighted).
2. **Key drivers for wheat:** High dietary reliance for protein, zinc, and iron, coupled with high heat exposure in South Asia and moisture deficits in North America and Russia.
3. **Maize and soybean** have high intrinsic heat sensitivity, but lower direct food importance globally because they are predominantly utilized for feed and industrial processing.
4. **Rice scores lowest** due to widespread irrigation and monthly mean temperatures below 35°C , though localized short-duration heat spikes during flowering remain an uncaptured risk.
5. **Iron** is the primary binding nutrient constraint across developing country diets.

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A Index Components and Scenario Data

Crop	Country	Heat		Water		Exposure		Sensitivity		I (%)	V_{F4}
		L (°C)	Δ (SD)	L (%)	Δ (SD)	E_h	E_w	S_h	S_w		
Rice	China (C)	0.00	0.77	0.0	0.00	0.00	0.00	0.17	0.50	13.3	0.000
	India (A)	0.00	1.05	1.3	0.00	0.00	0.02	0.66	0.50	15.6	0.002
	Bangladesh (A)	0.00	0.87	0.1	0.12	0.00	0.00	0.17	0.50	42.1	0.000
	Brazil (A)	0.00	1.08	7.6	0.06	0.00	0.14	0.17	0.50	4.7	0.004
	USA (C)	0.00	0.46	0.0	0.00	0.00	0.00	0.33	0.50	1.2	0.000
	Nigeria (A)	0.00	0.64	7.6	0.00	0.00	0.14	0.17	0.50	5.6	0.005
	Russia (C)	0.00	1.58	0.2	0.00	0.00	0.00	0.17	0.50	0.9	0.000
Wheat	China (C)	0.01	1.34	15.0	0.01	0.02	0.27	0.37	0.44	16.0	0.024
	India (C)	0.58	0.90	17.8	0.02	1.00	0.33	0.43	0.44	22.2	0.151
	Russia (D)	0.07	0.54	31.4	0.05	0.13	0.57	0.37	0.44	31.0	0.111
	USA (D)	0.08	0.24	36.9	0.01	0.13	0.68	0.37	0.44	18.2	0.075
	Egypt (B)	0.08	1.46	0.1	0.00	0.14	0.00	0.37	0.44	34.4	0.021
	Brazil (C)	0.01	0.96	7.3	0.25	0.02	0.13	0.37	0.44	13.3	0.010
Maize	USA (D)	0.01	0.38	16.3	0.00	0.02	0.30	0.46	0.50	2.0	0.004
	China (D)	0.01	0.77	8.5	0.00	0.01	0.15	0.40	0.66	1.3	0.002
	Brazil (A)	0.00	0.99	13.6	0.21	0.01	0.25	0.69	0.50	6.7	0.010
	India (A)	0.04	0.85	3.7	0.00	0.06	0.07	0.69	0.50	1.6	0.001
	S. Africa (B+M)	0.00	0.86	36.5	0.26	0.00	0.67	0.69	0.50	22.5	0.089
	Russia (D)	0.00	1.20	54.6	0.48	0.00	1.00	0.69	1.00	0.2	0.002
Soybean	Brazil (A)	0.00	1.08	3.8	0.13	0.00	0.07	0.76	0.42	7.3	0.002
	USA (D)	0.00	0.35	22.3	0.00	0.00	0.41	0.67	0.42	9.7	0.020
	Argentina (C)	0.00	0.55	26.1	0.19	0.00	0.48	0.76	0.61	1.5	0.005
	China (D)	0.00	0.71	13.8	0.00	0.00	0.25	0.76	0.56	5.8	0.010
	India (A)	0.00	0.59	0.9	0.00	0.00	0.02	0.76	0.42	2.4	0.000
	Russia (D)	0.00	0.63	32.5	0.09	0.00	0.60	0.76	0.42	1.1	0.003
	S. Africa (C+M)	0.00	0.83	35.4	0.33	0.00	0.65	0.76	0.42	3.8	0.012

Table 6: Detailed index components by crop and country.

Crop	Country	SSR	Climate-driven ΔY (%)		Standard -10% yield shock (S3)				
			S1 shift	S2 1-in-10	$\Delta kcal\%$	$\Delta Zn\%$	$\Delta Fe\%$	ΔMAR pts (no trade / trade)	
Rice	China	0.95	+0.0	+0.0	-2.36	-1.51	-0.82	+0.00 / +0.00	
	India	1.19	-4.8	-3.3	-2.47	-2.09	-1.15	-0.20 / -0.00	
	Bangladesh	1.02	+0.0	+0.0	-6.29	-5.45	-3.62	-0.50 / -0.51	
	Brazil	1.04	+0.0	+0.0	-0.78	-0.59	-0.40	-0.08 / +0.00	
	USA	1.43	-1.4	-2.7	-0.21	-0.15	-0.11	+0.00 / +0.00	
	Nigeria	1.00	+0.0	+0.0	-0.83	-0.76	-0.37	-0.07 / -0.07	
	Russia	0.79	+0.0	+0.0	-0.12	-0.09	-0.06	+0.00 / +0.00	
Wheat	China	0.90	-3.7	-4.0	-1.62	-1.92	-1.72	+0.00 / +0.00	
	India	1.01	-3.2	-4.1	-2.11	-3.26	-2.95	-0.52 / -0.28	
	Russia	1.78	-3.1	-4.8	-2.91	-4.56	-4.90	+0.00 / +0.00	
	USA	1.25	-1.1	-4.8	-1.54	-2.72	-3.09	+0.00 / +0.00	
	Egypt	0.56	-3.6	-3.3	-1.90	-2.70	-2.57	+0.00 / +0.00	
	Brazil	0.69	+1.1	-16.8	-0.75	-1.41	-1.57	-0.31 / +0.00	
Maize	USA	1.20	-2.8	-4.6	-0.21	-0.29	-0.33	+0.00 / +0.00	
	China	0.92	-2.8	-4.4	-0.15	-0.16	-0.14	+0.00 / +0.00	
	Brazil	1.55	-7.0	-6.1	-0.72	-0.92	-1.05	-0.20 / +0.00	
	India	1.15	-4.6	-2.7	-0.18	-0.23	-0.21	-0.04 / -0.00	
	S. Africa	1.36	-26.8	-23.8	-2.17	-2.95	-3.23	-0.53 / -0.00	
	Russia	1.28	-38.1	-31.4	-0.02	-0.03	-0.03	+0.00 / +0.00	
Soybean	Brazil	2.52	-4.3	-4.3	-0.98	-0.00	-0.01	-0.00 / +0.00	
	USA	1.65	-0.8	-3.9	-1.43	-0.00	-0.01	+0.00 / +0.00	
	Argentina	0.97	-4.6	-16.9	-0.20	-0.00	-0.00	+0.00 / +0.00	
	China	0.27	-2.6	-7.6	-0.10	-0.08	-0.18	+0.00 / +0.00	
	India	0.82	+0.8	-8.9	-0.17	-0.01	-0.03	-0.01 / -0.01	
	Russia	1.02	-4.4	-4.9	-0.13	-0.00	-0.01	+0.00 / +0.00	
	S. Africa	1.13	-7.1	-6.6	-0.43	-0.03	-0.09	-0.01 / -0.00	

Table 7: Yield-to-diet shock scenario results.